

Title of Your Analysis

A First Principles Exploration

Lim Jia Wei

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1. The Observation

Start here. Define the system you are looking at. Whether it's a centralized aircon unit or a lift algorithm, describe the real-world behavior that sparked this inquiry.

Hypothesis: Write a single sentence summarizing what you intend to prove or explore (e.g., “The inefficiency in lift scheduling comes from a lack of predictive modeling.”)

2. The Theoretical Model (Physics/Math)

This is where you show the “First Principles” depth. Break the problem down into equations.

We can model the system using the following relationship:

$$\mathcal{L} = \sum_i (T_{wait}^i + T_{travel}^i)$$

Explain the variables here. Why did you choose this model? What are the constraints?

3. Computational Analysis

Now, put on your Software Engineer hat. How do we simulate this? Insert your Python code below to model the theory.

```
#| echo: true #| label: fig-simulation #| fig-cap: “Simulation Results showing System Efficiency”
```

```
import matplotlib.pyplot as plt import numpy as np
```

1. Define the Parameters

```
x = np.linspace(0, 10, 100) noise = np.random.normal(0, 0.5, 100) y = 2 * x + 5 + noise # Theoretical model + noise
```

2. Plot the results

```
plt.figure(figsize=(6, 4)) plt.scatter(x, y, alpha=0.6, label='Observed Data') plt.plot(x, 2*x + 5, color='red',  
linestyle='-', label='Theoretical Ideal')
```

```
plt.title("Model vs Reality") plt.xlabel("Input Variable (e.g., Time)") plt.ylabel("Output Variable (e.g., Effi-  
ciency)") plt.legend() plt.grid(True, alpha=0.3) plt.show()
```